(μ, ϵ)-Robustness: Characterization, Existence, and Precision-Dependent Bounds

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Abstract. A classifier is ϵ-robust if it assigns the same label to any two inputs within distance ϵ. On connected domains such as $\mathbb{R}^n$, every ϵ-robust classifier is constant. Existing relaxations therefore weaken the guarantee to hand-picked or high-confidence inputs, or impose a bound on the probability of violations. Probabilistic bounds, however, constrain the probability of robustness violations, not where they lie. To bridge this gap, we introduce (μ, ϵ)-robustness, which requires every input to lie within the distance μ of a point with the same label at which the classifier is locally ϵ-robust. It coincides with ϵ-robustness at $\mu = 0$. We study (μ, ϵ)-robustness over semimetric spaces, which capture non-metric similarity notions used in fairness and natural language processing. Since deployed classifiers operate on finitely many representable inputs, we pay particular attention to finite domains such as IEEE 754 floating-point numbers. We characterize (μ, ϵ)-robustness by a covering condition that extends the classical equivalence between ϵ-robustness and separability, and we give necessary and sufficient conditions for the existence of a non-constant robust classifier. We derive bounds on ϵ and μ, in terms of the spacing between consecutive representable values, beyond which every (μ, ϵ)-robust classifier is constant; they hold on totally ordered domains and on partially ordered ones admitting a suitable decomposition. All results are mechanized in Lean 4: machine-checked proofs were developed with AI assistance; formal statements were validated against the paper by hand.

Keywords: Classifier Specification · Global Robustness · Finite-Precision Arithmetic · Semimetric Spaces · Neural Network Verification

1 Introduction

Machine learning classifiers can change their prediction under small input perturbations, raising concerns about safety, security, and ethics. A central notion for studying this phenomenon is *global robustness*: a classifier is ϵ-robust if it assigns

the same label to any two inputs within distance ϵ [5]. The distance encodes similarity and differs by domain. In computer vision, distances on pixel intensities are common [31]. In fairness, the distance encodes context-dependent similarity of individuals, for instance, being equally qualified for a job [6,18,26]. In natural language processing, it encodes syntactic and semantic similarity under meaningful linguistic variation [21]. Importantly, these notions of similarity may *not* be metric, as they may violate the triangle inequality [15,20]. We therefore study robustness over arbitrary *semimetric* spaces. On connected domains such as $\mathbb{R}^n$, every ϵ-robust classifier is essentially constant: overlapping ϵ-balls form chains between any two inputs, and the label must agree along each chain. Existing work has addressed this limitation by weakening the notion: local robustness restricts it to hand-picked inputs [23]; confidence-based notions to high-confidence inputs [2,19]; probabilistic notions bound the probability of drawing an ϵ-close pair with different labels [25,22]. Probabilistic bounds, however, constrain the probability of robustness violations, not their geometric structure: Fig. 1 shows a classifier that satisfies such a bound although no input is locally robust.

To address this limitation, we introduce (μ, ϵ)-robustness: Every input must lie within distance μ of a point with the same label at which the classifier is locally ϵ-robust. The property recovers ϵ-robustness at $\mu = 0$. Unlike probabilistic relaxations, it constrains where non-robust points may lie by imposing an explicit coverage structure on non-robust points: they form bands of width at most μ around locally robust points. Two of our main results, a covering characterization of robustness and a necessary-and-sufficient criterion for the existence of a non-constant robust classifier, hold over arbitrary semimetric spaces, providing a domain-independent foundation. Further, the impossibility result above is a consequence of *connected continuous* domains, whereas deployed classifiers compute over finite-precision arithmetic. In fact, verification methods reasoning over reals can be unsound due to their abstraction [16,8] showing that the fact that deployed systems operate over finite representable values rather than continuous domains has significant implications. On a finite domain, ϵ-balls can only build chains if ϵ exceeds the gaps between discrete values, so whether a non-constant robust classifier exists becomes a quantitative question about ϵ, μ, and the spacing between values of the domain.

Contributions. Studying (μ, ϵ)-robustness gives us domain-independent structural results and quantitative guarantees that can be specialized (see Section 4) to finite-precision arithmetic:

1. **Covering characterization.** We first introduce (μ, ϵ)-robustness (Section 2) and then present a covering characterization of (μ, ϵ)-robustness (Section 3), formally proving and extending the classical connection between ϵ-robustness and separability [34,24].
2. **Existence criteria.** We first introduce a floating-point model (Section 4), and then derive necessary and sufficient conditions for the existence of a non-constant robust classifier (Section 5), both for arbitrary semimetric spaces and for finite totally ordered domains.

3. **Precision-dependent bounds.** We obtain bounds on ϵ and μ in terms of the largest and smallest representable gaps (Section 6). These bounds explain empirically observed robustness gains under coarser quantization [32,27,12], and extend to some partially ordered domains.
4. **Mechanization.** All results are mechanized in Lean 4[4]. The machine-checked proofs were developed with AI assistance, and we manually validated that the formal statements match those in the document. To connect our floating-point model (Section 4) to an established IEEE 754 semantics, an AI assistant autonomously verified the constructions in Rocq on top of Flocq [4], and differentially tested the resulting executable distances against an independent implementation on special values and randomly generated binary 64-bit patterns.

Our results concern whether robust classifiers *exist* in a given domain; whether an *accurate* robust classifier exists and how robust models are trained and verified are orthogonal to our focus.

2 (μ, ϵ)-Robustness

We define classifier robustness in its usual form, survey related work containing other approaches to defining global robustness and highlight differences to our proposed formulation.

ϵ-Robustness. Local classifier robustness is an intuitive property widely accepted in the literature, which we formulate over semimetric spaces. We use the notation $[n] := \{1, 2, \ldots, n\} \subset \mathbb{N}$.

Definition 1 (Classifier). *Let S be a set of "inputs" and $[n]$ a set of "classes". A* classifier *is a total, surjective function: $c : S \to [n]$.*

Surjectivity is assumed to exclude unused labels and simplify notation.

Definition 2 (Semimetric Space). *Let (S, d) be a pair of a nonempty set and a function $d : S \times S \to \mathbb{R}_{\geq 0}$ such that for all $x, y \in S$, $d(x, y) = 0 \iff x = y$ (identity of indiscernibles) and $d(x, y) = d(y, x)$ (symmetry). Then, we call (S, d) a* semimetric space *and d a* distance *function on S.*

Using a distance, we define local ϵ-robustness at an input x^* requiring that sufficiently close (similar) inputs receive the same label. If it holds for all possible inputs, the classifier is globally ϵ-robust.

Definition 3 (ϵ-Robustness). *Let (S, d) be a semimetric space. A classifier $c : S \to [n]$ is* locally robust *at a point $x^* \in S$ for $\epsilon > 0$ if:*

$$\forall x \in S[d(x, x^*) \leq \epsilon \to c(x) = c(x^*)].$$

If a classifier c is given, we call $x \in S$ a *(non-)robust point* if c is (not) locally robust on x. If a classifier is robust at all points in S, it is *globally* ϵ-robust.

[4] https://github.com/verinncoq/mueps-robustness-proofs, archived in LAX as https://laxarchive.org/lax-544225/paper.html.

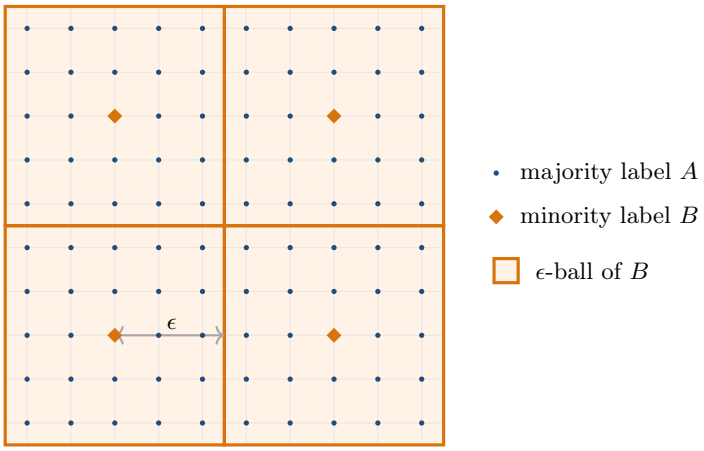


Fig. 1. Probabilistic robustness does not constrain the *structure* of non-robust points. In the image, inputs of class B are rare, so a randomly drawn ϵ-close pair is almost always same-label and a density-based robustness bound is met. Yet the B inputs sit on a sublattice whose ϵ-balls (which become squares with ℓ_∞ norm) cover the entire input domain: every input of the majority class A lies within ϵ of some input of class B, so no input is locally robust.

Research Gap. Consider the question of defining global robustness. An obvious way is to enforce local robustness for every point. But as ref. [23] points out, *"local robustness verification cannot be directly generalized to the whole decision region of a class"* because the distance can *"cross decision boundaries [...], making a verification task unrealistic"*. It is easy to see that points close to the true decision boundary are not robust which makes robustness on every point infeasible.

Many studies explore global robustness under additional assumptions. Different authors restrict robustness to specific datasets or underlying data distributions [5,30,35,34,24]. Some authors study the trade-off between robustness and accuracy and demonstrate that, for a particular classification problem, improvements in accuracy necessarily come at the cost of robustness [30,35]. In contrast, others show that robustness and accuracy can positively correlate when class distributions are separable, a property observed in several standard image datasets [34,24].

When it comes to measuring global robustness, many authors introduce additional concepts to avoid working with robustness at every point. A common approach is to enforce robustness only at points where a classifier has high confidence [2,19]. This solution creates another problem of ensuring that classifier prediction is achieved with high confidence whenever required [13,7]. Another popular approach is to enforce that the probability of finding a pair $x, x^* \in S$ with $d(x, x^*) \leq \epsilon$ and $c(x) \neq c(x^*)$ is less than some threshold value [22,23,25]. The probabilistic approach only bounds the density of robustness violations and does not tell anything about their structure. For example, a grid of singular

points such as one in Figure 1 with a different classifier label from its neighbors can result in a satisfaction of probabilistic robustness for a classifier that has no robust points (as they have a non-robust point from the grid as a neighbor).

This Work: (μ, ϵ)-Robustness. In this work, we explore a new version of global classifier robustness. We consider a classifier to be (μ, ϵ)-robust if every non-robust point has a robust point located at distance at most μ. Note that $(0, \epsilon)$-robustness is equivalent to ϵ-robustness.

Definition 4 ((μ, ϵ)-Robustness). *Let (S, d) be a semimetric space and $c : S \to [n]$ a classifier. c is (μ, ϵ)-robust for $\mu \in \mathbb{R}_{\geq 0}$ and $\epsilon \in \mathbb{R}_{>0}$ if for each $x \in S$, there exists $x' \in S$ with $d(x, x') \leq \mu$ such that $c(x) = c(x')$ and c is locally ϵ-robust at x'.*

This property is a novel domain-independent version of global classifier robustness. Intuitively, robust points act as anchors that cover the input domain. Non-robust points are allowed, but only if each of them lies within distance μ of a robust point with the same label. Thus, unlike probabilistic notions that only bound the number of robustness violations, (μ, ϵ)-robustness constrains their geometric arrangement. A non-robust region cannot appear arbitrarily far from robust regions but must be associated with a μ-close robust region. Conversely, only robust points can enable the existence of non-robust points within distance μ. In this sense, (μ, ϵ)-robustness relaxes global robustness without fully abandoning guarantees on the structure of robustness violations. In the following sections, we will explore properties of (μ, ϵ)-robust classifiers in more detail.

3 Robust Partitions via Covering

In this section, we pursue two goals. First, we introduce a formulation of (μ, ϵ)-robustness using partitions. We then show that (μ, ϵ) is equivalent to covering, mirroring equivalence of ϵ-robustness to separability.

3.1 Robust Partitions

To avoid ambiguities, we introduce some well-known definitions from the literature that are relevant to our developments.

Definition 5 (Partition). *A collection of nonempty sets $\{P_1, \ldots, P_p\}$ is a partition of a set S iff*

1. $\bigcup_{i \in [p]} P_i = S$ *and*
2. $P_i \cap P_j = \emptyset$ *for all* $i \neq j$.

The partition $\{S\}$ on the set S is called trivial.

Definition 6 (Classifier Induced Partition). *A classifier $c : S \to [n]$ induces a partition $P^c := \{S_1^c, S_2^c, \ldots S_n^c\}$ with $S_i^c := \{x \in S \mid c(x) = i\}$.*

Robustness is often described as invariance under ϵ-sized perturbations. This perspective allows us to reformulate robustness, for which we define the close neighborhood around an input as an ϵ-ball:

Definition 7 (ϵ-Ball). *Let (S, d) be a semimetric space. Then $B(x^*, \epsilon) := \{x \in S \mid d(x, x^*) \leq \epsilon\}$ is an* ϵ-ball *around x^*.*

Despite the name, $B(x^*, \epsilon)$ may not resemble a geometric ball. In fairness notions, for example, the distance function may capture similarity in qualification of individuals rather than geometric proximity of their attribute vectors.

Definition 8 ((μ, ϵ)-Robustness of a Partition). *Let (S, d) be a semimetric space. Then a partition $P := \{S_1, \ldots, S_p\}$ of S is* globally (μ, ϵ)-robust *for $\mu \geq 0, \epsilon > 0$ if and only if:*

$$\forall x \in S \, \exists x' \in B(x, \mu) \, [\bigvee_{i \in [p]} (\{x\} \cup B(x', \epsilon) \subseteq S_i)].$$

This subtle reformulation completely shifts (μ, ϵ)-robustness to a property of the induced partition. By working with partitions, our framework does not require the number of classes to be specified a priori, unlike some existing approaches [33]. To streamline subsequent results, we will use the term (μ, ϵ)-*robust partition* for partitions that satisfy Definition 8. If parameters are clear from the context, we will omit (μ, ϵ).

3.2 From Separability to Covering

A folklore characterization of global robustness is through its equivalence to ϵ-*separability* [34,24]. We can extend this property to our new robustness notion, albeit in a more complicated form as a covering condition.

Theorem 1. *Let (S, d) be a semimetric space, $P := \{S_1, \ldots, S_n\}$ a partition of S and $\mu \geq 0, \epsilon > 0$. Then the following are equivalent:*

1. *P is (μ, ϵ)-robust.*
2. *for all $S_i \in P$ and all $x \in S_i$ holds that $B(x, \mu) \not\subseteq \bigcup_{x^* \in S \setminus S_i} B(x^*, \epsilon)$.*

Proof. We prove both directions separately.

(1.) $\Rightarrow$ (2.): For contradiction, assume that (1.) holds but (2.) does not. In this case, there exists $S_i \in P$ with $x \in S_i$ such that $B(x, \mu) \subseteq \bigcup_{x^* \in S \setminus S_i} B(x^*, \epsilon)$. From Definition 8, there exists $x' \in B(x, \mu)$ with $B(x', \epsilon) \subseteq S_i$. From symmetry of distance (Definition 2), it follows that $x' \notin \bigcup_{x^* \in S \setminus S_i} B(x^*, \epsilon)$, leading to a contradiction with (1.).

(2.) $\Rightarrow$ (1.):

Similarly, assume that (2.) holds but not (1.). By negating the robustness condition in Definition 8, there exist $S_i \in P$ and $x \in S_i$ such that for every $x' \in B(x, \mu)$ we have $\{x\} \cup B(x', \epsilon) \not\subseteq S_i$. Since $x \in S_i$, there exist $x^* \in B(x', \epsilon)$ for each $x' \in B(x, \mu)$, such that $x^* \notin S_i$. By symmetry of distance, $x' \in B(x^*, \epsilon)$. This implies $B(x, \mu) \subseteq \bigcup_{x^* \in S \setminus S_i} B(x^*, \epsilon)$, raising a contradiction to (2.). $\square$

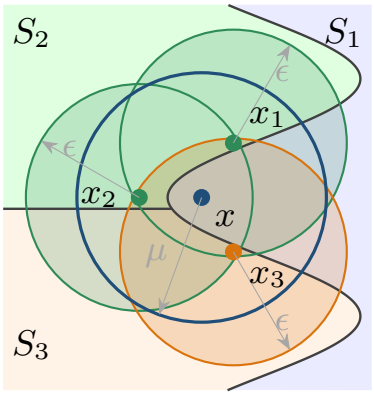


Fig. 2. Illustrated violation of (μ, ϵ)-robustness through covering condition (2.) of Theorem 1: the μ-ball of $x \in S_1$ is fully covered by three ϵ-balls of $x_1, x_2 \in S_2$ and $x_3 \in S_3$.

The condition (2.) in Theorem 1 can be stated as: no μ-neighbourhood is covered by ϵ-neighbourhoods of points with a different label. An example when it is not the case is illustrated in Figure 2. As (μ, ϵ)-robustness is a generalization of ϵ-robustness, our proof is also a formalization of separability for ϵ-robustness. For $\mu = 0$, the condition (2.) reduces to each point (0-neighbourhood) being ϵ-far from any point with a different label. Separability is established here in a general, non-probabilistic setting without distributional assumptions. Unlike prior work, we show that separability is a *necessary* condition for global robustness.

The characterization of robustness from Theorem 1 is central to future results and we will use it as a primary criterion of partition robustness.

4 Modeling Finite Precision Arithmetic

To bridge theory and practice, we want to formulate our results on a structure that exists under computer arithmetic constraints. Modeling classifiers over real numbers can lead to a mismatch between theoretical insights and practical implementations. For example, verifiers for neural networks based on real arithmetic can be unsound when applied to floating-point computations [16,8].

Mathematical Model. Floating point numbers approximate real values with finite precision according to the IEEE 754 standard, which defines a finite arithmetic over a set of bitvector-encoded numbers. Reasoning about finite-precision arithmetic with standard operators (such as addition) is challenging, since these operators may violate familiar algebraic properties like associativity [17]. An additional problem is presented by special floating point values such as `NaN` or `Inf`. Nevertheless, the IEEE 754-2008 standard provides a well-defined total or-

dering on floating-point numbers [28], implemented, for example, in Rust[5]. This motivates the formulation of our results on ordered spaces $(S, <, d)$.

However, an introduction of an order will not lead to more productive results unless a relationship between the order and the distance exists. To fix this, we introduce a new inequality. The inequality states that visiting an intermediate point between two points in the order produces at most the direct distance. In some sense, it is the opposite of a triangle inequality, which states that visiting an additional point increases the distance. But, as our goal in further proofs is to bound the size of balls, we found this inequality more productive.

With these considerations, we arrive at our mathematical model standing for computer arithmetic spaces:

Definition 9 (Ordered Semimetric Space). *A tuple $(S, <, d)$ with a set S, a (partial) order $<\subseteq S \times S$, and $d : S \times S \to \mathbb{R}_{\geq 0}$ is an* ordered semimetric space *if (S, d) is a semimetric space and for all $a, b, c \in S$ such that $a \leq b \leq c$ implies $d(a, c) \geq d(a, b) + d(b, c)$.*

For example, $(\mathbb{R}, \leq, d)$ is an ordered semimetric space for $d(x, y) = |x - y|$ and $d(x, y) = (x - y)^2$. Note that with the second distance, $\mathbb{R}$ does not form a metric space.

Model Validation. After defining ordered semimetric spaces, the question remains whether such spaces indeed exist on floating point numbers and can resemble common distances. To verify this, we employed autonomous AI-driven proof engineering using Mistral Vibe and GLM-5.3. We prompted it to generate provably correct instances of ordered semimetric spaces with respect to the IEEE 754 total order on lists of Flocq [4] floating point numbers within the Rocq proof assistant [29], re-implement the resulting distances for Python 64-bit `float` and produce a script for differential testing between Rocq and Python using fixed edge cases and randomly generated instances. The AI assistant generated the artifacts with minimal human intervention. The output was reviewed by humans later and is available on GitHub[6] for further independent inspection. The following proposition summarizes the ordered semimetric constructions identified through this AI-assisted model validation process.

Proposition 1. *Let $\mathcal{F}$ be a binary floating point format with $p \in \mathbb{N}$ being the number of mantissa bits. Then, it consists of triples (s, c, m) representing numbers $(-1)^s \cdot 2^c \cdot m$ such that $s \in \{0, 1\}$, $c \in \{c_{\min}, \ldots, c_{\max}\} \subset \mathbb{Z}$, $m \in \{k \cdot 2^{-p} : k \in \{1, \ldots, 2^{p+1} - 1\} \subset \mathbb{Z}\}$ together with additional values ± 0, `NaN`, $\pm$`Inf` $\in \mathcal{F}$. For each $(s, c, m) \in \mathcal{F}$ holds either $m \geq 1$ (normal) or $c = c_{\min}$ (subnormal). Let $\phi : \mathcal{F} \to \mathbb{R}$ with:*

- $\phi((s, c, m)) = (-1)^s \cdot 2^c \cdot m$
- $\phi(+0) = 0$ *and* $\phi(-0) = -2^{c_{\min} - p - 1}$

[5] https://doc.rust-lang.org/std/primitive.f64.html#method.total_cmp

[6] https://github.com/verinncoq/mueps-robustness-proofs/tree/main/flocq_ordered_spaces

- $\phi(\texttt{NaN}) = 2^{c_{\max}+1} + 1$
- $\phi(\pm\texttt{Inf}) = \pm 2^{c_{\max}+1}$

Then, the following are ordered semimetric spaces:

1. $(\mathcal{F}, <_{IEEE}, d_1)$ *where* $<_{IEEE}$ *is the IEEE total order and* $d_1(f_1, f_2) = |\phi(f_1) - \phi(f_2)|$.
2. $(\mathcal{F}, <_{IEEE}, d_2)$ *where* $d_2(f_1, f_2) = (\phi(f_1) - \phi(f_2))^2$.
3. $(\mathcal{F}^n, <^n, d_3(f, g) := \sum_{i\in[n]} d(f_i, g_i))$ *such that* $(\mathcal{F}, <, d)$ *is an ordered semimetric space and* $<^n$ *is n-wise product partial order.*
4. $(\mathcal{F}^n, <^n_*, d_4(f, g) := \max_{i\in[n]} d(f_i, g_i))$ *where* $(\mathcal{F}, <, d)$ *is an ordered semimetric space such that for all* $a, b, c \in \mathcal{F}$ *with* $a \leq b \leq c$ *holds that* $d(a, c) = d(a, b) + d(b, c)$ *and* $<^n_* := \{(f, g) \in \mathcal{F}^n \times \mathcal{F}^n \,|\, f <^n g \wedge \forall i, j \in [n]\, d(f_i, g_i) = d(f_j, g_j)\}$.
5. *Extensions of (3) and (4) to arbitrary finite lists of floating-point numbers where a shorter list is padded with* `NaN` *for distance computation, and a penalty equal to* 1 *is added to the distance whenever list lengths are different. The partial orders compare only lists of the same length, and lists of different lengths are incomparable.*

Note that Proposition 1 was verified for the second time in Lean as a part of the AI-assisted formalization. For simplicity, the floating-point model uses a single canonical positive-sign `NaN` rather than all IEEE 754 `NaN` encodings. The construction can be extended to the full IEEE 754 total order by assigning distinct, order-preserving real values to the remaining `NaN` encodings. The combinations of spaces 1 and 3, 1 and 4 in Proposition 1 give us notions similar to l_1, l_∞ distances, respectively. Overall, we showed that ordered semimetric spaces can be faithfully constructed on floating point numbers, showing that the results proven in the following sections are transferrable to computer arithmetic.

5 Existence of Robust Partitions

The main criticism of ϵ-robustness is that it is impossible to satisfy as separability cannot be enforced on a decision boundary. We will explore our two-fold answer to this critique. First, we should notice that the non-satisfiability argument only applies to continuous domains such as real numbers, while most classifiers operate on a discrete domain of floating-point numbers. Second, this problem is alleviated for (μ, ϵ)-robustness. To support two arguments with mathematical evidence, we will prove conditions for existence of at least one non-constant robust classifier partition. In Section 5.1, we derive a general criterion on an arbitrary semimetric space, and in Section 5.2, we specialize it to ordered semimetric spaces, thus creating a connection to finite precision arithmetic.

5.1 General Existence Criterion

We study the conditions under which (μ, ϵ)-robustness is satisfiable by a non-constant classifier, i.e. one that distinguishes between at least two classes. The

following result shows that (μ, ϵ)-robustness imposes structural constraints on the domain.

Theorem 2 (General Existence). *Let (S, d) be a semimetric space and $\mu \geq 0, \epsilon > 0$. The following statements are equivalent:*

1. *The only $(\mu, \epsilon)-$robust partition on S is trivial.*
2. *For all nonempty $S' \subset S$, there exists $x \in S$ such that either $x \in S'$ and $B(x, \mu) \subseteq \bigcup_{x' \in S \setminus S'} B(x', \epsilon)$ or $x \in S \setminus S'$ with $B(x, \mu) \subseteq \bigcup_{x' \in S'} B(x', \epsilon)$.*

Proof. We prove each direction separately:

(1.) $\Rightarrow$ (2.): Assume (1.) and let $S' \subset S$ be a proper subset such that condition (2.) is false. We prove that $P := \{S', S \setminus S'\}$ is (μ, ϵ)-robust, leading to contradiction with (1.). If P is not robust, then by Theorem 1 there exists $x \in S$ with either $x \in S'$ and $B(x, \mu) \subseteq \bigcup_{x' \in S \setminus S'} B(x', \epsilon)$ or $x \in S \setminus S'$ with $B(x, \mu) \subseteq \bigcup_{x' \in S'} B(x', \epsilon)$. Note that it is exactly the condition (2.) of this theorem, which was assumed to be false, implying by contradiction that P is a robust partition.

(2.) $\Rightarrow$ (1.): Assume (2.) and let $P := \{S_1, \ldots, S_n\}$ with $n > 1$ be any robust partition of S. For contradiction, we show that S_1 violates (2.). By condition (2.) from Theorem 1, there is no $x \in S_1$ with $B(x, \mu) \subseteq \bigcup_{x' \in S \setminus S_1} B(x', \epsilon)$. It leaves the case that $x \in S \setminus S_1$ with $B(x, \mu) \subseteq \bigcup_{x' \in S_1} B(x', \epsilon)$. This means that $x \in S_i$ for some $i \in \{2, \ldots, n\} \subset [n]$ and by Theorem 1, $B(x, \mu) \not\subseteq \bigcup_{x' \in S \setminus S_i} B(x', \epsilon)$. As $\bigcup_{x' \in S_1} B(x', \epsilon) \subseteq \bigcup_{x' \in S \setminus S_i} B(x', \epsilon)$, S_1 violates condition (2.) of this theorem in all cases. □

Given fixed μ and ϵ, Theorem 2 establishes a class of semimetric spaces that have at least one nontrivial robust partition. It shows that impossibility of robustness is not a universal phenomenon but depends on structural properties of the underlying domain. In fact, many sets and distance functions violate condition (2) of Theorem 2. For example, consider the set $\{0, 2, 4, 6\}$ with the standard metric $d(x, y) = |x - y|$ and let $\epsilon = 1$. This observation demonstrates the benefits of providing mathematical clarity by replacing informal arguments with rigorous analysis.

5.2 Existence of a Robust Partition on a Total Order

Denote by $Succ(x, y)$ a formula that is true if $x < y$ and there is no z with $x < z < y$. We start with results on finite totally ordered semimetric spaces $(S, <, d)$ where S is arbitrary and the order $<$ is total. For each subset $S' \subseteq S$ of such a space, a *boundary* $\partial S' = \{x \in S' \mid \exists y \notin S'. Succ(x, y) \vee Succ(y, x)\}$ and *complement boundary* $\overline{\partial} S' = \{x \notin S' \mid \exists y \in S'. Succ(x, y) \vee Succ(y, x)\}$.

Theorem 3. *Let $(S, <, d)$ be a finite totally ordered semimetric space and $\mu \geq 0, \epsilon > 0$. The following two statements are equivalent:*

1. *For all nonempty $S' \subset S$, there exists $x \in S$ such that either $x \in S'$ and $B(x, \mu) \subseteq \bigcup_{x' \in S \setminus S'} B(x', \epsilon)$ or $x \in S \setminus S'$ with $B(x, \mu) \subseteq \bigcup_{x' \in S'} B(x', \epsilon)$.*

2. *For all nonempty $S' \subset S$, there exists $x \in S$ such that either $x \in S'$ and $(B(x,\mu) \cap S') \subseteq \bigcup_{x' \in \overline{\partial} S'} B(x', \epsilon)$ or $x \in S \setminus S'$ with $(B(x,\mu) \cap (S \setminus S')) \subseteq \bigcup_{x' \in \partial S'} B(x', \epsilon)$.*

Proof. Again, we prove each direction separately:

(1.) ⇒ (2.): To prove (2.) of this theorem, pick any nonempty $S' \subset S$. By applying (1.) to S', we get two cases:

- *1. Case: $x \in S'$ and $B(x,\mu) \subseteq \bigcup_{x' \in S \setminus S'} B(x', \epsilon)$.* To show (2.), we will prove that $(B(x,\mu) \cap S') \subseteq \bigcup_{x' \in \overline{\partial} S'} B(x', \epsilon)$. Pick some $x^* \in B(x,\mu) \cap S'$ and select some $x'' \in S \setminus S'$ with $x^* \in B(x'', \epsilon)$, which exists by assumption. W.l.o.g. assume that $x^* < x''$, implying existence of a $x_{\overline{\partial}} \in \overline{\partial} S'$ with $x^* \leq x_{\overline{\partial}} \leq x''$. Then, $d(x^*, x_{\overline{\partial}}) \leq d(x^*, x'') \leq \epsilon$ by Definition 9. From this, $x^* \in B(x_{\overline{\partial}}, \epsilon)$ follows. As such $x_{\overline{\partial}} \in \overline{\partial} S'$ exists for each $x^* \in B(x,\mu) \cap S'$, we deduce $(B(x,\mu) \cap S') \subseteq \bigcup_{x' \in \overline{\partial} S'} B(x', \epsilon)$, and (2.) is proven.
- *2. Case: $x \in S \setminus S'$ and $B(x,\mu) \subseteq \bigcup_{x' \in S'} B(x', \epsilon)$.* Proven analogously to the 1. Case by finding $x_\partial \in \partial S'$ for each $x^* \in B(x,\mu) \cap S \setminus S'$ for some $x \in S$.

(2.) ⇒ (1.): To prove (1.), pick a nonempty $S' \subset S$. By (2.), there exists $x \in S$ satisfying one of two conditions in (2.). We get two cases:

- *1. Case: $x \in S'$ and $(B(x,\mu) \cap S') \subseteq \bigcup_{x' \in \overline{\partial} S'} B(x', \epsilon)$.* Consider some $x^* \in B(x,\mu)$. If $x^* \in S'$ then by (2.), $x^* \in \bigcup_{x' \in \overline{\partial} S'} B(x', \epsilon) \subseteq \bigcup_{x' \in S \setminus S'} B(x', \epsilon)$. If $x^* \in S \setminus S'$ then $x^* \in B(x^*, \epsilon) \subseteq \bigcup_{x' \in S \setminus S'} B(x', \epsilon)$. Combined, we deduce $B(x,\mu) \subseteq \bigcup_{x' \in S \setminus S'} B(x', \epsilon)$ and (1.) is proven.
- *2. Case: $x \in S \setminus S'$ and $(B(x,\mu) \cap (S \setminus S')) \subseteq \bigcup_{x' \in \partial S'} B(x', \epsilon)$.* Proven similarly to the 1. Case.

□

Corollary 1. *Let $(S, <, d)$ a finite totally ordered semimetric space and $\mu \geq 0, \epsilon > 0$. The following two statements are equivalent:*

1. *The only robust partition on S is trivial.*
2. *For all nonempty $S' \subset S$, there exists $x \in S$ such that either $x \in S'$ and $(B(x,\mu) \cap S') \subseteq \bigcup_{x' \in \overline{\partial} S'} B(x', \epsilon)$ or $x \in S \setminus S'$ with $(B(x,\mu) \cap (S \setminus S')) \subseteq \bigcup_{x' \in \partial S'} B(x', \epsilon)$.*

Proof. Note that condition (1.) in Theorem 3 is exactly condition (2.) in Theorem 2.

With this, we established necessary and sufficient conditions for the existence of a nontrivial robust partition in an ordered semimetric space. Note that since Theorem 3 is an equivalence condition, condition (2.) cannot be simplified further. A covered $B(x,\mu)$ is not necessarily produced by $x \in \partial S'$ or $x \in \overline{\partial} S'$. In addition, multiple ϵ-balls may be required to cover a μ-ball. In summary, we can observe that studying equivalent conditions to existence does not lead to simple conditions. Due to this observation, we will switch to estimating bounds on parameters μ and ϵ for nontrivial robust partitions.

6 Precision-Related Upper Bounds

An interesting empirically supported fact for ϵ-robustness is that quantization, a process of reducing numerical precision of classifier's inputs and parameters, causes a classifier to become more robust [32,27,12]. The explanation is simple: larger distances between input space points raises the threshold on admissible ϵ-values and thereby makes ϵ-robustness easier to satisfy. Mathematically, one can show that with larger gaps, the maximum ϵ for nontrivial robust partitions increases. In this section, our goal is to derive such bounds on μ and ϵ based on precision measures: in Section 6.1 for totally ordered semimetric spaces and in Section 6.2 for general partial orders such as multidimensional inputs common to classifiers such as neural networks.

6.1 Upper Bounds on Totally Ordered Spaces

We start with upper bounds on totally ordered semimetric spaces introduced in Section 5.2. To state our results, we define characteristics of distances between two distinct representable numbers in an order representing finite-precision arithmetic. For a totally ordered semimetric space $(S, <, d)$, $a, b \in S$ with $a \leq b$, define an *interval* $[a, b] := \{a \leq x \leq b \,|\, x \in S\}$.

Definition 10 (Precision Gap Metrics). *Let $(S, <, d)$ be a finite totally ordered semimetric space with $|S| \geq 2$. We define the metrics Δ and δ as:*

$$\Delta := \max(\{\frac{d(x,y)}{|[x,y]| - 1} \,|\, x, y \in S \wedge x < y\})$$
$$\delta := \max(\{\frac{|[x,y]| - 1}{d(x,y)} \,|\, x, y \in S \wedge x < y\})$$

where $|\cdot|$ is set cardinality.

As an example, consider the set of representable floating-point values of a binary format, where -0, $\pm$`Inf` and `NaN` are excluded, and the distance $d(x, y) = |x - y|$. In this case, Δ equals the largest gap between consecutive representable values, the maximal ULP of the format, and δ is the reciprocal of the smallest such gap, the minimal (subnormal) ULP [11]. Now, we show how Δ and δ define upper bounds on ϵ and necessary lower bounds on μ for nontrivial robust partitions.

Theorem 4. *Let $(S, <, d)$ a finite totally ordered semimetric space with $|S| \geq 2$ and $\mu \geq 0, \epsilon > 0$. If $\epsilon \geq \Delta \cdot \delta \cdot \mu + \Delta$ then for all $x, x' \in S$ with $Succ(x, x')$, it holds that $B(x, \mu) \subseteq B(x', \epsilon)$.*

Proof. First, observe that $B(x, \mu) \subseteq B(x', \epsilon)$ can be restated as: for all $x^* \in S$ with $d(x^*, x) \leq \mu$ holds $d(x^*, x') \leq \epsilon$. Now, fix any $x^*, x, x' \in S$ with $Succ(x, x')$. If $x^* = x$ then $d(x^*, x') = d(x, x') \leq \Delta \leq \epsilon$. If $x^* = x'$, $d(x^*, x') = 0 \leq \epsilon$. We are left with two cases:

– *1. Case:* $x^* < x < x'$:

$$\begin{aligned} d(x^*, x') =& \frac{d(x^*, x')}{|[x^*, x']| - 1} \cdot (|[x^*, x']| - 1) \leq \Delta \cdot (|[x^*, x']| - 1) = \\ & \Delta \cdot |[x^*, x]| = \Delta \cdot (|[x^*, x]| - 1) + \Delta = \\ & \Delta \cdot \frac{|[x^*, x]| - 1}{d(x^*, x)} \cdot d(x^*, x) + \Delta \leq \Delta \cdot \delta \cdot d(x^*, x) + \Delta \leq \\ & \Delta \cdot \delta \cdot \mu + \Delta \leq \epsilon \end{aligned}$$

– *2. Case:* $x < x' < x^*$:

$$\begin{aligned} d(x^*, x') =& d(x', x^*) \leq d(x, x^*) \\ & \leq \mu = \frac{d(x, x')}{d(x, x')} \cdot \mu \leq \Delta \cdot \delta \cdot \mu < \epsilon \end{aligned}$$

□

Corollary 2. *Let $(S, <, d)$ a finite totally ordered semimetric space with $|S| \geq 2$ and $\mu \geq 0, \epsilon > 0$. If $\epsilon \geq \Delta \cdot \delta \cdot \mu + \Delta$ then the only (μ, ϵ)-robust partition is trivial.*

Proof. We show that conclusion of the Theorem 4 implies condition (2.) of Theorem 3. Fix a nonempty $S' \subset S$. Let $x_{min} := min(S \setminus S')$ and $x'_{min} := min(S')$. Since neither $S \setminus S'$ nor S' are empty, both x_{min} and x'_{min} exist.

– *1. Case:* $x_{min} < x'_{min}$. Let $x_{pre} := max(\{x \in S \setminus S' \mid x < x'_{min}\})$ and $x'_{succ} := x'_{min}$. As $Succ(x_{pre}, x'_{succ})$ holds, $B(x_{pre}, \mu) \subseteq B(x'_{succ}, \epsilon)$ by Theorem 4. Observe that $x'_{succ} \in \partial S'$. Condition (2.) of Theorem 3 holds as $x_{pre} \in S \setminus S'$ and $(B(x_{pre}, \mu) \cap S \setminus S') \subseteq B(x_{pre}, \mu) \subseteq B(x'_{succ}, \epsilon) \subseteq \bigcup_{x' \in \partial S'} B(x', \epsilon)$.
– *2. Case:* $x_{min} > x'_{min}$. Let $x'_{pre} := max(\{x' \in S' | x' < x_{min}\})$ and $x_{succ} := x_{min}$, then $Succ(x'_{pre}, x_{succ})$. Similarly to the 1. Case, $(B(x'_{pre}, \mu) \cap S') \subseteq B(x'_{pre}, \mu) \subseteq B(x_{succ}, \epsilon) \subseteq \bigcup_{x' \in \overline{\partial} S'} B(x', \epsilon)$.

Corollary 1 states that condition (2.) is equivalent to $(S, <, d)$ having the trivial partition as the only (μ, ϵ)-robust partition.

For ϵ-robustness ($\mu = 0$), we see that $\epsilon \geq \Delta$ is an upper bound for a nontrivial robust partition. With increasing Δ, this bound also increases which provides theoretical evidence for better robustness after quantization. Introduction of a new parameter μ allows to relax strong restrictions imposed by ϵ on partitions, which is the goal of having a weaker version of robustness.

6.2 Upper Bounds on Partial Orders

Usually, classifiers are not defined on a single floating-point number line but on multi-dimensional vectors that potentially consist of numbers with different precision (e.g. integers and floats). The total order given by an arithmetic over each vector dimension creates a partial order over the entire vector space. This multi-dimensional setting demands an extended result on the achievable robustness of a classifier. Unfortunately, (μ, ϵ)-robustness shows non-compositional behavior which we want to illustrate with a lemma:

Lemma 1. *Given a semimetric space (S, d) let $S_1, S_2 \subset S$ with $S_1 \cap S_2 \neq \emptyset$ as well as $\mu \geq 0, \epsilon > 0$. The following statements are **false**:*

1. *If the only (μ, ϵ)-robust partition on S_1 is trivial, all (μ, ϵ)-robust partitions P on S include a superset of S_1.*
2. *If the only (μ, ϵ)-robust partitions on S_1 and S_2 are trivial, the only (μ, ϵ)-robust partition on $S_1 \cup S_2$ is trivial.*

Proof. We provide counterexamples for $S := \{a, b, c, d\}$ with $d(a, b) = d(b, c) = d(c, d) = 1$, $d(a, c) = d(b, d) = 2$ and $d(a, d) = 3$. Note that S has a nontrivial robust partition for $\mu = \epsilon = 1$: $\{\{a, b\}, \{c, d\}\}$.

1. $S_1 = \{b, c\}$ does not have a nontrivial robust partition for $\mu = \epsilon = 1$, and the nontrivial robust partition on S does not contain a superset of S_1.
2. $S_1 = \{a, b, c\}$ and $S_2 = \{b, c, d\}$ do not have a nontrivial robust partition for $\mu = \epsilon = 1$, but S does.

□

Still, it is possible to find a general but relatively weak decomposition principle. First, we define how the decomposing structure looks like:

Definition 11 (μ-Decomposition). *A collection of nonempty sets $D := \{D_1, \ldots, D_p\}$ is a μ-decomposition of a semimetric space (S, d) iff*

1. *for each $x \in D_i$, $B(x, \mu) \subseteq D_i$ and*
2. *for each $x_1, x_2 \in S$, there is a sequence of sets $D_{i_1}, \ldots, D_{i_n} \in D$ such that $D_{i_j} \cap D_{i_{j+1}} \neq \emptyset$ and $x_1 \in D_{i_1}$, $x_2 \in D_{i_n}$.*

Now, we are ready to show that μ-decompositions allow us to focus on considering (μ, ϵ)-robustness of individual decomposition elements only.

Theorem 5. *Given a semimetric space (S, d) with $\mu \geq 0, \epsilon > 0$, let D be a μ-decomposition of S. If each $D_i \in D$ has the trivial partition as its only (μ, ϵ)-robust partition then the only (μ, ϵ)-robust partition on S is trivial.*

Proof. We prove the condition (2.) from Theorem 2 directly in two steps. Pick a nonempty $S' \subset S$. In the first step, we rule out existence of some S' and in the second, construct the witness $x \in S$ that satisfies Theorem 2 (2.).

(*Step 1*): First, we prove that there exists $D_i \in D$ such that both $D_i \cap S'$ and $D_i \cap (S \setminus S')$ are nonempty. For contradiction, assume that $D_i \cap S' = \emptyset$ or $D_i \cap (S \setminus S') = \emptyset$ for all $D_i \in D$. As S' is a proper nonempty subset, there are $x' \in S'$ and $x^* \in S \setminus S'$. Using property (2.) from Definition 11, construct a sequence $D_{i_1}, \ldots, D_{i_p}$ out of elements in D with $x' \in D_{i_1}$ and $x^* \in D_{i_p}$. As $D_{i_1} \subseteq S'$ and $D_{i_p} \cap S' = \emptyset$ by assumption of this step, there is a consecutive pair $D_{i_j}, D_{i_{j+1}}$ with $j \in [p]$ in the sequence such that $D_{i_j} \cap S' = D_{i_j}$ and $D_{i_{j+1}} \cap S' = \emptyset$. But, by property (2.) of Definition 11, $D_{i_j} \cap D_{i_{j+1}} \neq \emptyset$, leading to contradiction: $\emptyset \neq D_{i_j} \cap D_{i_{j+1}} = (D_{i_j} \cap S') \cap D_{i_{j+1}} = D_{i_j} \cap (S' \cap D_{i_{j+1}}) = \emptyset$.

(*Step 2*): We construct the witness $x \in S$ in this step. Per Step 1, pick D_i with $D_i \cap S'$ and $D_i \cap (S \setminus S')$ being both nonempty. Then, as D_i does not have

nontrivial robust partitions by theorem assumption, by Theorem 2, there exists $x \in D_i$ such that either $x \in D_i \cap S'$ and $(B(x,\mu) \cap D_i) \subseteq \bigcup_{x' \in D_i \setminus S'} (B(x',\epsilon) \cap D_i)$ or $x \in D_i \cap (S \setminus S')$ with $(B(x,\mu) \cap D_i) \subseteq \bigcup_{x' \in D_i \cap S'} (B(x',\epsilon) \cap D_i)$. We pick x as the witness for our proof. If $x \in S'$ then by property (1.) of Definition 11, $(B(x,\mu) \cap D_i) = B(x,\mu) \subseteq \bigcup_{x' \in D_i \setminus S'} (B(x',\epsilon) \cap D_i) \subseteq \bigcup_{x' \in S \setminus S'} B(x',\epsilon)$. Similarly if $x \in S \setminus S'$, $(B(x,\mu) \cap D_i) = B(x,\mu) \subseteq \bigcup_{x' \in D_i \cap S'} (B(x',\epsilon) \cap D_i) \subseteq \bigcup_{x' \in S'} B(x',\epsilon)$. As in both cases, condition (2.) from Theorem 2 is proven, and S does not have a nontrivial robust partition in this case. □

We specialize the result from Theorem 5 to ordered semimetric spaces.

Corollary 3. *Let $(S, <, d)$ a finite partially ordered semimetric space with $|S| \geq 2$ and $\mu \geq 0, \epsilon > 0$. Let $D := \{D_1, \ldots, D_n\}$ be a μ-decomposition of S such that all $D_i \in D$ form a totally ordered semimetric space $(D_i, <, d)$ with corresponding Δ_i, δ_i. For singleton sets D_i, we define $\Delta_i = \delta_i = 0$. If $\epsilon \geq \max_{i \in [n]} (\Delta_i \cdot \delta_i \cdot \mu + \Delta_i)$ then the only (μ, ϵ)-robust partition of S is trivial.*

Proof. First, observe that each $D_i \in D$ with $|D_i| \geq 2$, Corollary 2 implies that D_i does not have a nontrivial (μ, ϵ)-robust partition, for $|D_i| = 1$ this holds trivially. This implies by Theorem 5 that every (μ, ϵ)-robust partition on S is trivial.

Corollary 3 presents an upper bound for any partially ordered set that has a μ-decomposition. Despite our motivation, a general product order does not necessarily admit such decomposition. This means that the Corollary does not easily generalize to vectors of finite precision numbers.

7 Conclusion

Discussion. This work addresses the key limitation of global ϵ-robustness: its stringent requirements on classifiers. As clearly visible in Theorem 4, the parameter μ increases the upper bound on ϵ in totally ordered semimetric spaces such as floating-point numbers (Section 5.2). (μ, ϵ)-robustness is proven equivalent in Theorem 1 to a coverage condition in parallel to ϵ-robustness being equivalent to separability. This leads to necessary and sufficient conditions for the existence of a non-constant (μ, ϵ)-classifier in the general case (Theorem 2) and for totally ordered semimetric spaces (Corollary 1). Precision-dependent upper bounds (Corollaries 2 and 3) create a connection between the underlying arithmetic and the existence of non-constant classifiers. Setting $\mu = 0$, one obtains well-known results on ϵ-robustness that were, to the best of our knowledge, never collected and formally proven in a single place.

More generally, the results show that robustness is not solely a property of a classifier but also of the structure of its input domain relative to a chosen distance. By relating robustness to covering and the spacing of representable values, our study of (μ, ϵ)-robustness produced exact statements that demonstrate how the choice of domain and distance constrains the existence of robust

classifiers. This perspective is relevant for neural networks, which operate on finite-precision arithmetic rather than continuous domains, a mismatch between which complicates proofs of soundness for verifiers [16,8].

Future work. The next logical step is to design a verification algorithm for (μ, ϵ)-robustness. This is a significant open research problem, because the (μ, ϵ)-robustness property is in the class of $\forall\exists$-hyperproperties. Although these are studied in general context [9,10,14], neural network verification currently focuses on the domain of hyperproperties with $\forall$ quantifiers only [2,1,3].

AI Use Disclosure

In addition to the uses mentioned in the text, AI tools were also used to draft and edit the text of the article, simulate peer review, and generate LaTeX code for the figures. The tools used were Mistral Vibe, Microsoft Copilot, Overleaf AI assistant, as well as Claude Code and ChatGPT. After using these tools, the authors reviewed and edited the content as needed and assume full responsibility for the content of the publication.

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